

IoT-Based Smart City Waste Monitoring and Collection Optimization Platform

Project Report

Smart City Waste Management using IoT, ESP32, Ultrasonic Sensors, Python Analytics and ThingSpeak Cloud

1. Executive Summary

This project presents an IoT-based Smart City Waste Monitoring and Collection Optimization Platform. The system continuously monitors waste-bin fill levels using ultrasonic sensing, analyzes collection requirements, generates alerts, and visualizes data on the ThingSpeak cloud dashboard. The platform helps municipalities optimize collection schedules and reduce operational costs.

2. Problem Statement

Traditional waste collection systems follow fixed schedules, resulting in unnecessary collection trips, fuel wastage, and overflowing bins. This project provides a real-time monitoring solution that enables data-driven waste collection decisions.

3. System Architecture

Ultrasonic Sensor → ESP32/Simulation → Fill-Level Calculation → Priority Engine → ThingSpeak Cloud → Dashboard → Analytics & Reporting

4. Technologies Used

- ESP32 Microcontroller
- Ultrasonic Sensor (HC-SR04)
- Python Programming
- Pandas Analytics
- ThingSpeak Cloud Platform
- ReportLab PDF Generation
- CSV Data Logging

5. ThingSpeak Dashboard

The dashboard provides real-time visualization of waste-bin status through:

- Fill Percentage Gauge
- Distance Monitoring
- Overflow Alert Indicator
- Collection Required Indicator
- Historical Fill-Level Trends
- Collection Request Trends

6. KPI Analysis

Average Fill Level : 46.73%
Maximum Fill Level : 94.16%
Minimum Fill Level : 0.27%
Total Alerts : 0
Collection Requests : 14
Total Records Logged : 141

7. Collection Priority System

HIGH Priority Bins : 13
MEDIUM Priority Bins : 12
LOW Priority Bins : 20

The priority engine automatically categorizes waste bins and assists municipal authorities in scheduling waste collection operations.

8. Predictive Waste Collection

Increase waste collection frequency.

9. Business Benefits

- Reduces unnecessary collection trips
- Prevents waste overflow
- Reduces fuel consumption
- Improves operational efficiency
- Supports smart-city initiatives
- Enables data-driven decision making

10. Future Scope

- Integration with real ESP32 hardware
- GPS-enabled smart bins
- Route optimization using AI
- Mobile application for operators
- Machine Learning based waste prediction
- Integration with municipal smart-city platforms

Report Generated Automatically